PB-7339-M04G-10

硬件测试报告

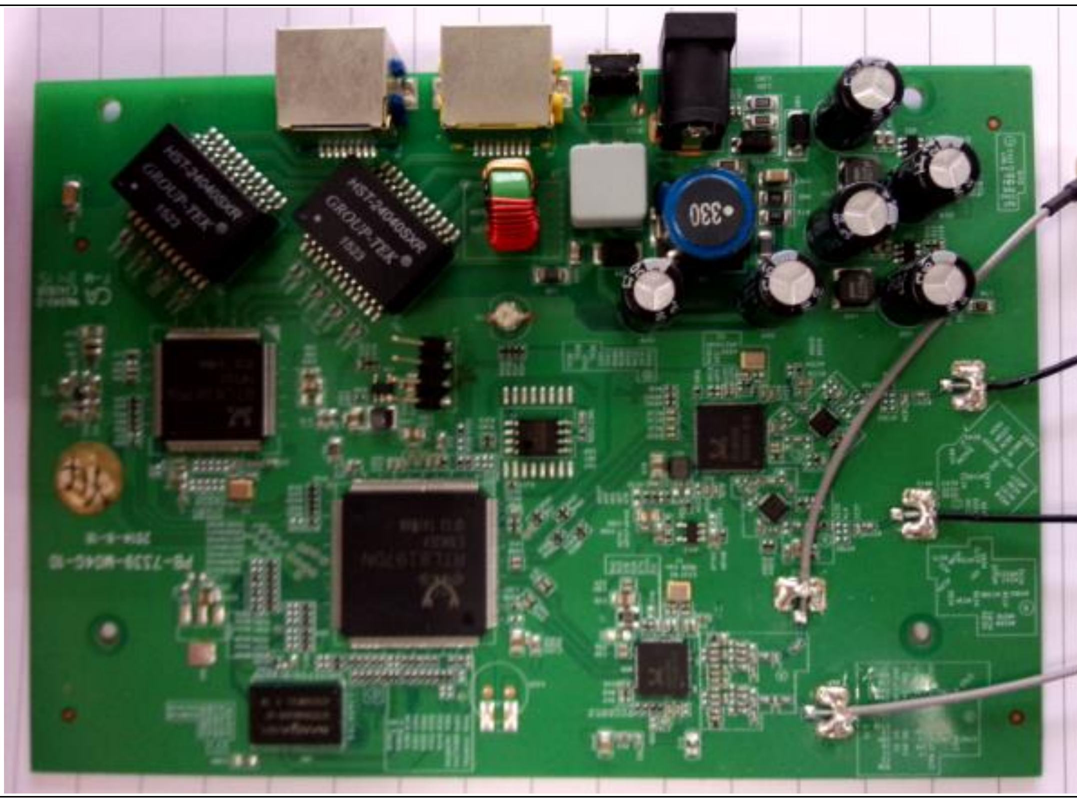
测试工程师：汪节

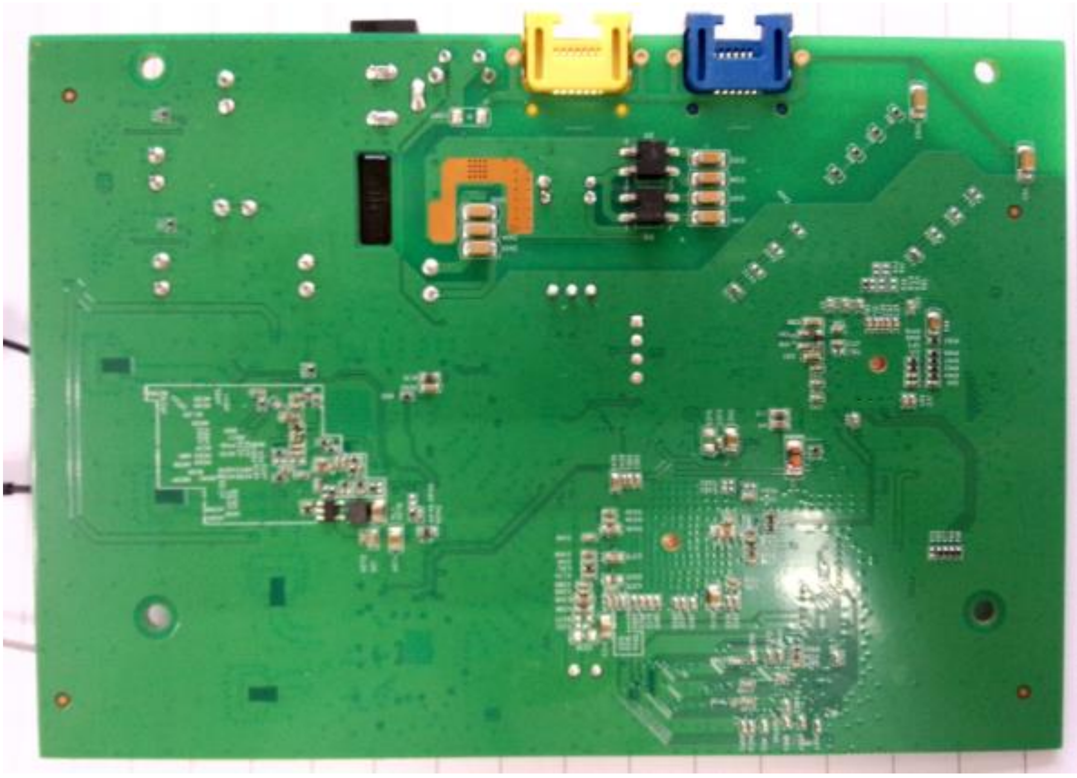
审核：王朋刚

核准：刘松辉

目录

一、测试基本信息

PCB 版本编号	PB-7339-M04G-10
机构编号	7339
芯片方案	RTL8197DN+RTL8812AR+RTL8192ER
测试时间	2015-9-18
测试样机规格	2G+5G 2T2R 标准速率 300Mbps+867Mbps wireless AP/Router
测试样机软件版本	\\192.168.32.112\开发一部\04-产品软件\开发用软件\APROUTER\7339\7339-0915.IMG
测试样机板数量	18pcs
测试地点	开发一部实验室
测试设备	IQflex，示波器，万用表 ，屏蔽箱，高温箱
附加说明	管理地址：192.168.1.254 产测程序: WiFiTest for RtlAp - 1.10.103.640
产品图片	



二、测试项目及结果

测试项目			是否测试	测试结果
电源	AC-DC	输入输出电压	/	/
		输入电压范围	/	/
		输入输出电流	/	/
		输入输出纹波	/	/
		输入输出功率	/	/
	DC-DC	输入输出电压	✓	✓
		输入电压范围	✓	✓
		输入输出电流	✓	✓
		输入输出纹波	✓	✓
		输入输出功率	✓	✓
信号测量	上电顺序	顺序	✓	✓
	时钟	频率	✓	✓
		幅度	✓	✓
		抖动	✓	✓
	RF	CCK RX（灵敏度）	✓	✓
		CCK TX(POW、EVM、频偏)	✓	✓

		OFDM RX（灵敏度）	✓	✓
		OFDM TX(POW、EVM、频偏)	✓	✓
		MCS20 TX（灵敏度）	✓	✓
		MCS20 TX(POW、EVM、频偏)	✓	✓
		MCS20 TX（灵敏度）	✓	✓
		MCS20 TX(POW、EVM、频偏)	✓	✓
	数据信号	各接口数据	✓	✓
基本功能	串口	端口速率，可操作性	✓	✓
	按键	作用是否有效	✓	✓
	指示灯	亮的方式	✓	✓
性能	无线性能	无线吞吐量测试	✓	✓
		室内覆盖测试	✓	✓
	有线端口交换		✓	✓
环境测试	烧机		✓	✓
	冷启动测试		✓	✓
	DUT 高低温功能及温度		✓	✓
上下大电测试	上下电测试		✓	✓

三、测试记录

1、电源

1.1、DC TO DC

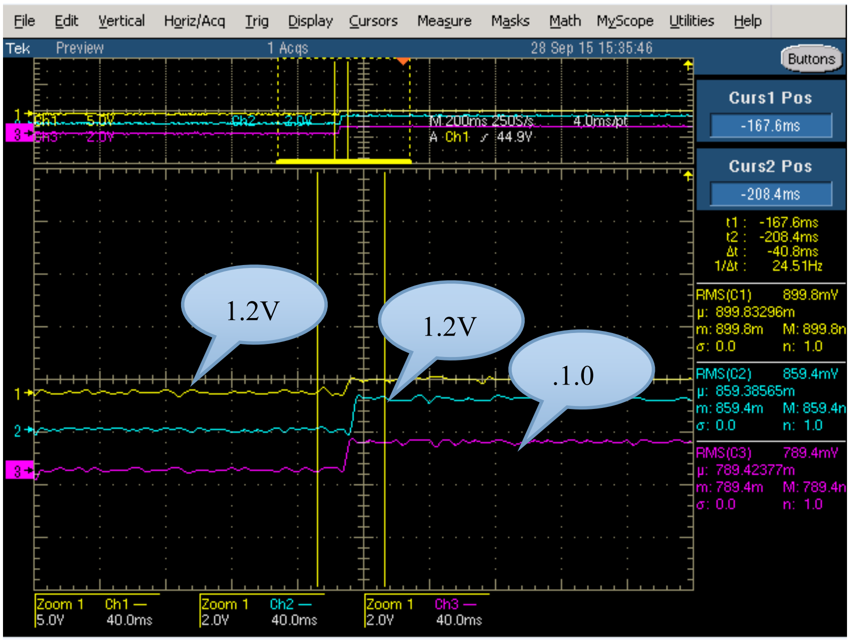
DC 12V/1A 供电与标准 POE 供电

空载				满载			
电压(V)	最大/平均电流(A)	纹波(mV)	平均功率(W)	电压(V)	最大/平均电流(A)	纹波(mV)	平均功率(W)
48V (POE)	0.152/0.108	/	5.184	48V (POE)	0.244/0.155	/	7.44
12.03V (DC)	0.472/0.335	/	4.03	12.03V(DC)	0.832/0.529	/	6.363
5.114V	0.328/0.041	97	0.209	5.114V	0.640/0.147	112	0.751
3.4V	1.4/0.873	54	2.968	3.4V	2.2/1.56	96	5.304
1.293V	0.284/0.230	68	0.297	1.293V	0.360/0.235	74	0.303
1.219V	0.244/0.156	/	0.190	1.219V	0.592/0.219	/	0.266
1.066V	0.980/0.706	42	0.752	1.066V	1.14/0.936	51	0.941

2、信号测量

2.1、数字信号

<table>
<tr><th>信号名称</th><th>频率</th><th>图片</th></tr>
<tr><td>上电顺序</td><td></td><td>

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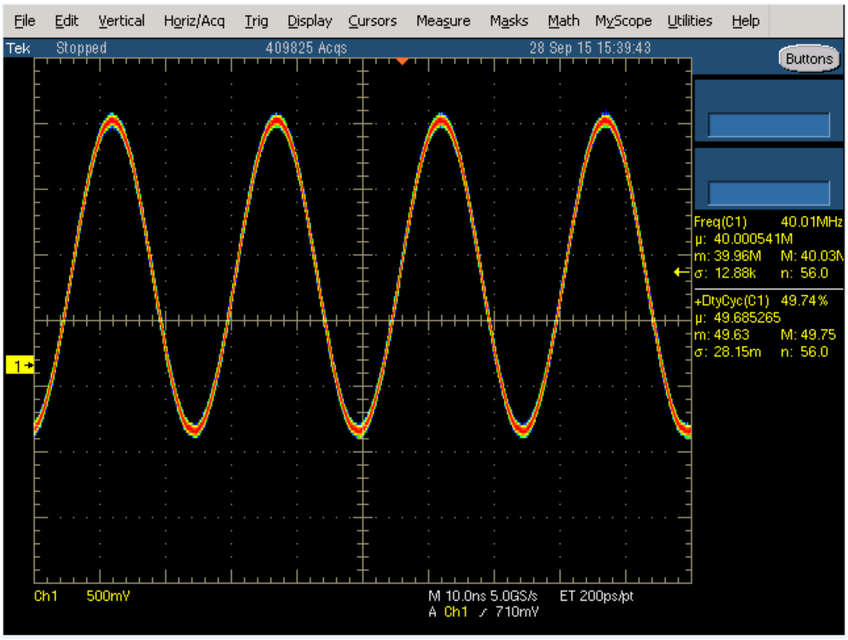
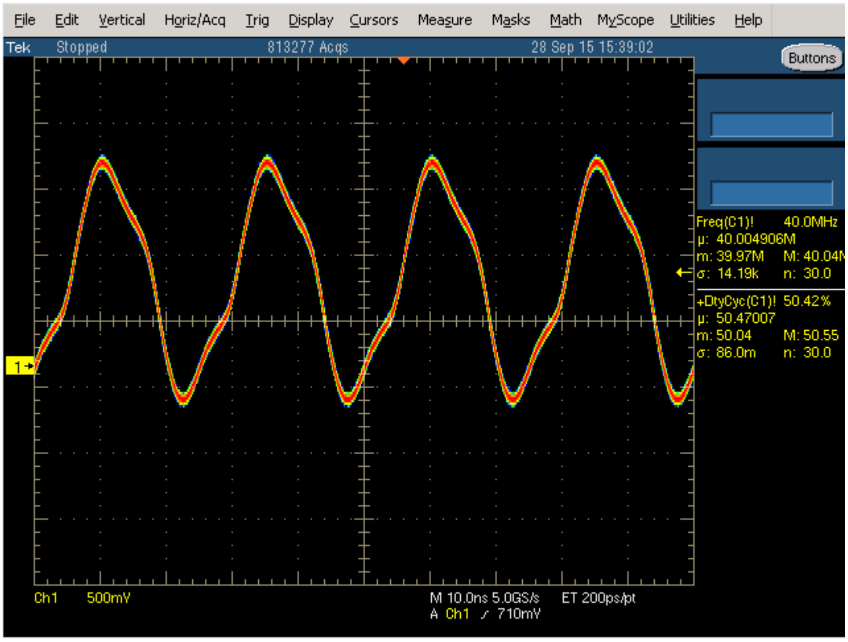
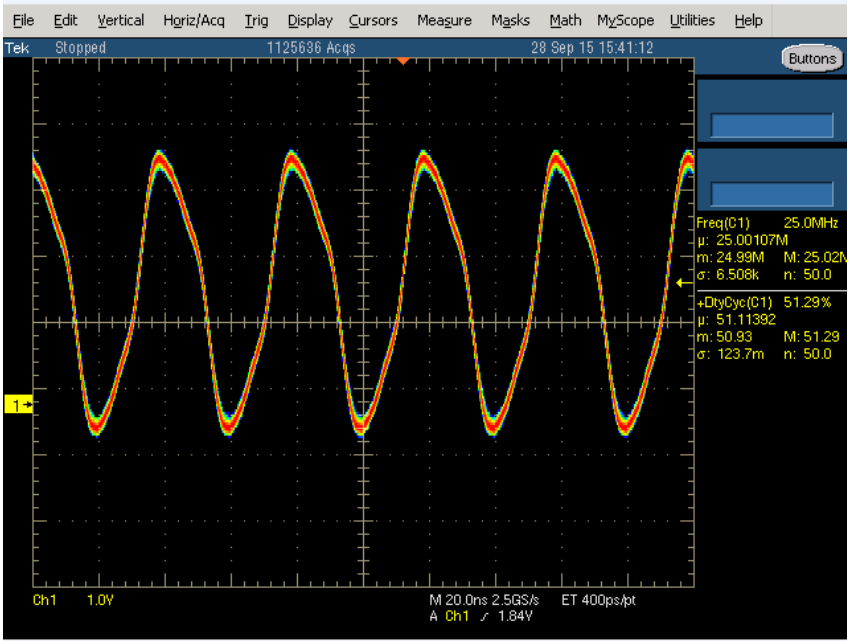
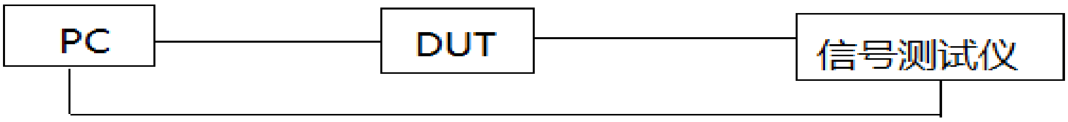
2G 晶振	40.0MHZ	
5G 晶振	40.0MHZ	

	晶振	25.0MHZ	

2.2、wifi 信号

1）RF 测量

拓扑结构



测试记录

5G- TX

antenna	A		datarate	OFDM-54						
channel	36	52	64	100	120	140	149	153	165	169
Power	+21.59	+21.11	+21.26	+20.26	+19.70	+20.27	+20.37	+20.61	+20.61	+21.45
Evm	-26.14	-27.29	-25.05	-25.37	-25.96	-26.44	-26.53	-25.74	-27.05	-26.53
ppm	-1.12	-1.2	-1.22	-1.25	-1.24	-1.26	-1.22	-1.24	-1.26	-1.26
Mask	0	0	0	0	0	0	0	0	0	0
antenna	A		datarate	HT20-MCS7						
channel	36	52	64	100	120	140	149	153	165	169
Power	+20.90	+20.44	+20.07	+19.37	+19.23	+19.71	+19.80	+20.04	+20.08	+20.92
Evm	-28.42	-29.35	-29.7	-28.31	-28.19	-28.83	-28.22	-28.36	-28.79	-28.77
ppm	-1.28	-1.28	-1.27	-1.26	-1.25	-1.27	-1.27	-1.24	-1.27	-1.27
Mask	0	0	0	0	0	0	0	0	0	0
antenna	A		datarate	HT40-MCS7						
channel	38	46	62	102	118	134	151	159		
Power	+21.49	+21.24	+20.67	+19.87	+19.63	+20.27	+20.40	+20.37		
Evm	-28.33	-28.73	-29.38	-28.66	-29.1	-28.24	-28.14	-28.8		
ppm	-1.24	-1.22	-1.23	-1.22	-1.21	-1.2	-1.19	-1.21		
Mask	0.29	0.1	0	0.03	0.07	0.13	0.16	0.03		

antenna	A		datarate	VHT20-S1-MCS8						
channel	36	52	64	100	120	140	149	153	165	169
Power	+20.98	+20.49	+20.14	+19.44	+19.26	+19.75	+19.86	+20.12	+20.17	+20.36
Evm	-32.1	-32.43	-32.64	-32.34	-32.24	-32.39	-32.51	-32.07	-32.3	-32.64
ppm	-1.24	-1.24	-1.25	-1.24	-1.23	-1.25	-1.24	-1.23	-1.25	-1.26
Mask	0	0	0	0	0	0	0	0	0	0
antenna	A		datarate	VHT40-S1-MCS9						
channel	38	46	62	102	118	134	151	159		
Power	+19.99	+20.14	+19.77	+18.70	+18.30	+18.80	+18.88	+19.30		
Evm	-32.82	-32.26	-32.26	-32.53	-32.88	-32.89	-32.81	-32.26		
ppm	-1.22	-1.21	-1.19	-1.2	-1.19	-1.17	-1.1	-1.15		
Mask	0	0	0	0	0	0	0	0		
antenna	A		datarate	VHT80-S1-MCS9						
channel	42	58	106	122	138	155				
Power	+19.68	+19.79	+18.44	+18.63	+18.89	+18.94				
Evm	-32.8	-32.26	-32.1	-32.37	-32.37	-32.55				
ppm	-1.16	-1.17	-1.18	-1.16	-1.2	-1.15				
Mask	0	0	0	0	0	0				

antenna	B		datarate	OFDM-54						
channel	36	52	64	100	120	140	149	153	165	169
Power	+21.54	+21.05	+20.90	+19.59	+19.62	+19.69	+20.06	+19.89	+20.71	+21.56
Evm	-26.45	-26.65	-25.02	-26.63	-25.33	-26.7	-25.95	-26.82	-25.22	-25.21
ppm	+0.27	+0.20	+0.16	+0.14	+0.15	+0.14	+0.13	+0.12	+0.14	+0.13
Mask	0	0	0	0	0	0	0	0	0	0

antenna	B		datarate	HT20-MCS7					
channel	36	52	64	100	120	140	153	165	169
Power	+21.02	+20.47	+19.54	+19.00	+18.59	+19.51	+19.40	+19.79	+20.63
Evm	-28.47	-28.31	-30.35	-29.03	-28.71	-28.35	-28.74	-28.59	-28.23
ppm	+0.09	+0.09	+0.07	+0.10	+0.08	+0.10	+0.10	+0.13	+0.13
Mask	0	0	0	0	0	0	0	0	0

antenna	B		datarate	HT40-MCS7				
channel	38	46	62	102	118	134	151	159
Power	+21.39	+21.29	+20.72	+19.60	+19.41	+19.51	+20.12	+19.84
Evm	-28.05	-28.39	-28.64	-28.52	-28.08	-29.72	-28.11	-29.21
ppm	+0.09	+0.12	+0.11	+0.09	+0.09	+0.09	+0.11	+0.09
Mask	0.2	0.23	0.07	0.03	0.07	0	0.1	0

antenna	B		datarate	VHT20-S1-MCS8					
channel	36	52	64	100	120	140	153	165	169
Power	+21.08	+20.48	+19.59	+19.01	+18.63	+19.53	+19.41	+19.85	+20.68
Evm	-32.15	-33.25	-34.26	-33.75	-33.81	-32.48	-32.98	-32.12	-32.13
ppm	+0.10	+0.07	+0.08	+0.06	+0.05	+0.06	+0.05	+0.06	+0.05
Mask	0	0	0	0	0	0	0	0	0

antenna	B		datarate	VHT40-S1-MCS9				
channel	38	46	62	102	118	134	151	159
Power	+19.93	+19.94	+19.15	+18.09	+18.13	+18.40	+18.57	+18.85
Evm	-32.47	-32.22	-33.55	-33.8	-32.24	-32.64	-33.27	-32.23
ppm	+0.06	+0.09	+0.09	+0.10	+0.09	+0.10	+0.12	+0.09
Mask	0	0	0	0	0	0	0	0

antenna	B		datarate	VHT80-S1-MCS9	
channel	42	58	106	122	138
Power	+19.86	+19.21	+18.30	+18.01	+18.80
Evm	-32.61	-33.9	-32.31	-33.37	-32.01
ppm	+0.08	+0.08	+0.08	+0.08	+0.10
Mask	0	0	0	0	0

5G-RX

A	36	38	42	46	52	58	62	64	100	102	106	118	120	122	134	138	140	149	151	153	155	159	165	169
OFDM-54	-17/-76				-17/-75			-17/-75	-17/-75				-17/-76				-17/-76	-17/-76		-17/-76			-17/-76	-18/-77
OFDM-6	-17/-91				-17/-91			-17/-91	-17/-92				-17/-92				-17/-92	-17/-92		-17/-92			-17/-92	-18/-92
HT20-MCS7	-17/-73				-17/-73			-17/-73	-17/-73				-17/-73				-17/-73	-17/-73		-17/-73			-17/-73	-18/-74
HT20-MCS0	-17/-91				-17/-91			-17/-91	-17/-92				-17/-91				-17/-92	-17/-92		-17/-92			-17/-91	-18/-92
HT40-MCS7		-17/-70		-17/-70			-17/-70			-17/-70		-17/-70			-17/-70				-17/-70			-17/-70		
HT40-MCS0		-17/-89		-17/-89			-17/-89			-17/-89		-17/-89			-17/-89			-17/-89	-17/-89			-17/-89		
VHT20-S1-MCS8	-17/-68				-16/-68			-17/-68	-17/-68				-17/-67				-17/-68	-17/-68		-17/-68			-17/-68	-18/-68
VHT20-S1-MCS0	-17/-91				-16/-91			-17/-91	-17/-91				-17/-91				-17/-92	-17/-92		-17/-92			-17/-91	-18/-92
VHT40-S1-MCS9		-17/-64		-17/-64			-17/-63			-17/-63		-17/-63			-17/-63				-17/-63			-17/-63		
VHT40-S1-MCS0		-17/-89		-17/-89			-17/-89			-17/-89		-17/-89			-17/-89			-17/-89	-17/-89			-17/-89		
VHT80-S1-MCS9			-17/-60			-17/-59					-17/-59			-17/-59		-17/-59					-17/-59			
VHT80-S1-MCS0			-17/-86			-17/-86					-17/-86			-17/-86		-17/-86					-17/-86			

B	36	38	42	46	52	58	62	64	100	102	106	118	120	122	134	138	140	149	151	153	155	159	165	169
OFDM-54	-17/-76				-17/-75			-17/-75	-17/-76				-17/-76				-17/-76	-17/-76		-17/-76			-17/-76	-18/-77
OFDM-6	-17/-91				-17/-91			-17/-91	-17/-91				-17/-91				-17/-91	-17/-92		-17/-92			-17/-92	-18/-92
HT20-MCS7	-17/-73				-17/-73			-17/-73	-17/-73				-17/-73				-17/-73	-17/-73		-17/-73			-17/-73	-18/-74
HT20-MCS0	-17/-91				-17/-91			-17/-91	-17/-91				-17/-91				-17/-91	-17/-91		-17/-91			-17/-91	-18/-92
HT40-MCS7		-17/-70		-17/-70			-17/-70			-17/-70		-17/-70			-17/-70				-17/-70			-17/-70		
HT40-MCS0		-17/-89		-17/-89			-17/-89			-17/-89		-17/-89			-17/-89			-17/-89	-17/-89			-17/-89		
VHT20-S1-MCS8	-17/-68				-17/-68			-17/-68	-17/-68				-17/-68				-17/-68	-17/-68		-17/-68			-17/-68	-18/-68
VHT20-S1-MCS0	-17/-91				-17/-91			-17/-91	-17/-91				-17/-91				-17/-91	-17/-91		-17/-91			-17/-91	-18/-92
VHT40-S1-MCS9		-17/-63		-17/-63			-17/-63			-17/-63		-17/-62			-17/-63				-17/-63			-17/-62		
VHT40-S1-MCS0		-17/-89		-17/-88			-17/-89			-17/-89		-17/-89			-17/-89			-17/-89	-17/-89			-17/-89		
VHT80-S1-MCS9			-17/-58			-17/-59					-17/-58			-17/-58		-17/-58					-17/-58			
VHT80-S1-MCS0			-17/-86			-17/-86					-17/-86			-17/-86		-17/-86					-17/-86			

2.4G-TX:

antenna	A		datarate	CCK-11
channel	1	7	13	
Power	+25.01	+24.94	+24.95	
Evm	-24.87	-24.75	-26.36	
ppm	+1.53	+1.53	+1.24	
Mask	27.95	23.53	17.08	

antenna	A		datarate	OFDM-54
channel	1	7	13	
Power	+19.57	+19.82	+19.86	
Evm	-25.61	-25	-25.22	
ppm	+1.54	+1.60	+1.62	
Mask	0	0	0	

antenna	A		datarate	HT20-MCS7
channel	1	7	13	
Power	+19.27	+19.04	+19.19	
Evm	-28.04	-28.72	-28.39	
ppm	+1.62	+1.68	+1.66	
Mask	0	0	0	

antenna	A		datarate	HT40-MCS7
channel	3	7	11	
Power	+18.62	+18.44	+18.79	
Evm	-28.4	-29.11	-28.63	
ppm	+1.63	+1.62	+1.62	
Mask	0	0	0	

antenna	B		datarate	CCK-11
channel	1	7	13	
Power	+24.19	+24.32	+24.48	
Evm	-25.63	-27.59	-27.94	
ppm	+1.58	+1.40	+1.36	
Mask	29.06	32.74	32.54	

antenna	B		datarate	OFDM-54
channel	1	7	13	
Power	+19.58	+19.49	+19.75	
Evm	-25.1	-25.92	-25.94	
ppm	+1.58	+1.65	+1.68	
Mask	0	0	0	

antenna	B		datarate	HT20-MCS7
channel	1	7	13	
Power	+18.97	+19.25	+19.44	
Evm	-28.78	-28.61	-28.38	
ppm	+1.70	+1.68	+1.68	
Mask	0	0	0	

antenna	B		datarate	HT40-MCS7
channel	3	7	11	
Power	+19.06	+18.80	+18.92	
Evm	-28.16	-28.89	-28.82	
ppm	+1.66	+1.65	+1.65	
Mask	0.07	0	0	

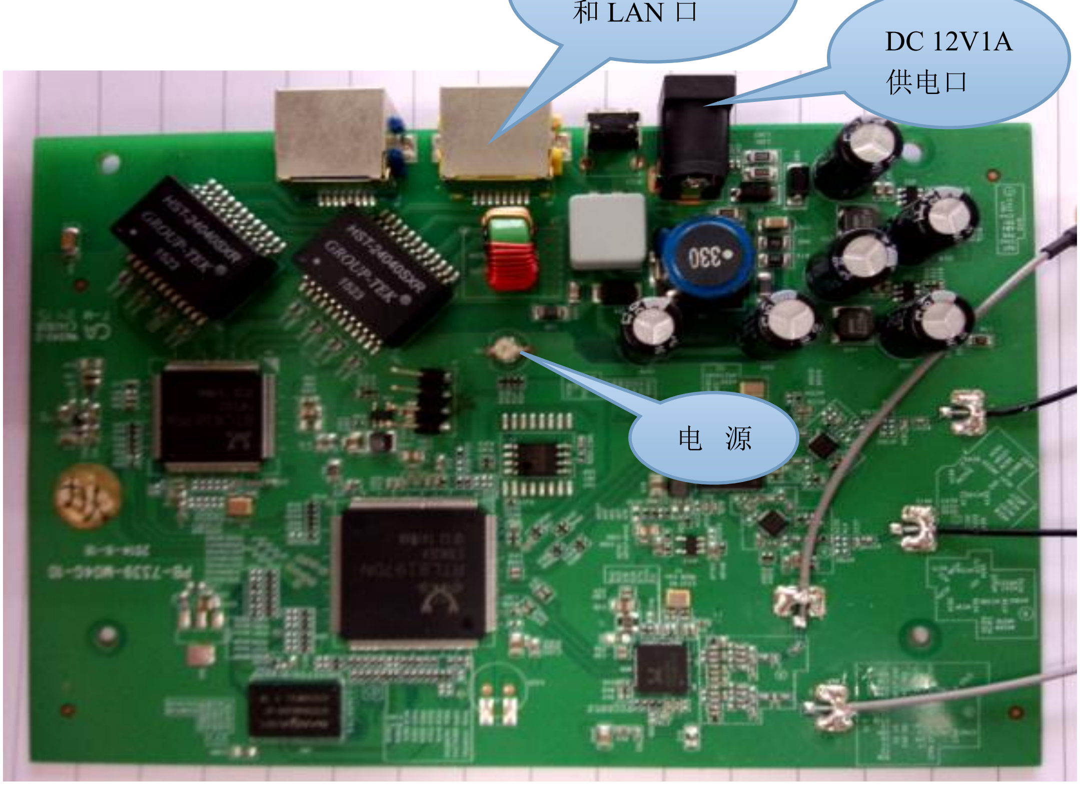
2.4G-RX:

A	1	3	7	11	13
CCK-1	-11/-94		-11/-94		-11/-95
CCK-11	-11/-86		-11/-86		-11/-86
OFDM-6	-16/-91		-16/-91		-16/-89
OFDM-54	-16/-74		-16/-74		-16/-73
HT20-MCS0	-16/-90		-16/-90		-16/-89
HT20-MCS7	-16/-71		-16/-71		-16/-70
HT40-MCS0		-16/-87	-16/-87	-16/-87	
HT40-MCS7		-16/-68	-16/-67	-16/-67	

B	1	3	7	11	13
CCK-1	-11/-94		-11/-95		-11/-94
CCK-11	-11/-86		-11/-86		-11/-86
OFDM-6	-16/-90		-16/-90		-16/-91
OFDM-54	-16/-74		-16/-73		-16/-73
HT20-MCS0	-16/-89		-16/-89		-16/-89
HT20-MCS7	-16/-71		-16/-71		-16/-70
HT40-MCS0		-16/-87	-16/-87	-16/-87	
HT40-MCS7		-16/-67	-16/-67	-16/-67	

3、基本功能

3.1、LED 指示灯



3.2、串口

串口速率：38400 测试 OK。

3.3、 硬件按键

复位键功能：OK

4、性能测试

4.1、无线吞吐量测试

2．4G

陪测网卡：RTL8812AU

2.4G	40M 270M			20M 130M		
28#磊科软件	TX+RX	TX	RX	TX+RX	TX	RX
1	170.116	175.859	150.262	94.015	93.450	89.268
7	183.540	172.582	173.511	95.126	93.904	89.570
11	185.180	175.409	175.811	95.545	93.829	89.045

2.4G	40M 300M			20M 144.5M		
4#原厂软件	TX+RX	TX	RX	TX+RX	TX	RX
1	187.291	188.670	187.257	105.502	104.606	99.221
7	188.701	188.291	190.680	105.272	104.803	99.451
11	189.840	189.049	182.220	105.142	104.657	99.366

2.4G	40M			20M		
16#原厂软件	TX+RX	TX	RX	TX+RX	TX	RX
1	187.531	187.830	190.671	100.746	104.392	90.414
7	178.317	187.499	172.219	103.919	103.723	96.275
11	190.915	189.331	190.497	104.869	104.288	97.715

2.4	40M			20M		
原厂软件	TX+RX	TX	RX	TX+RX	TX	RX
27#	189.781	189.011	184.291	104.624	104.371	97.990
14#	187.538	188.689	173．635	105.372	98.621	98.536
11#	187.988	189.424	181.398	105.435	104.558	99.090
24#	189.451	189.118	174.258	102.727	103.507	96.292
磊科软件						
2#	183.047	164.258	177.284	109.417	106.092	106.707
22#	184.134	165.425	183.338	90.220	87.528	87.697
29#	184.293	168.181	182.874	106.059	99.654	104.211
12#	184.428	163.605	176.412	108.682	104.287	105.727
21#	182.893	162.571	177.254	89.063	86.490	87.803
17#	178.964	161.449	180.837	88.827	87.483	86.887

5G

磊科软件									
5G	80M780M			40M 360M			20M 130M		
28#	TX+RX	TX	RX	TX+RX	TX	RX	TX+RX	TX	RX
36	218.744	224.023	200.662	190.531	190.558	198.942	106.577	98.306	106.366
149	218.184	222.298	201.062	216.856	207.612	200.803	103.178	105.734	97.355
161	216.784	217.395	201.062	199.242	214.773	198.803	99.721	105.577	94.265
原厂软件									
5G	80M 867M			40M 400M			20M 173.5M		
4#	TX+RX	TX	RX	TX+RX	TX	RX	TX+RX	TX	RX
36	241.214	259.866	216.264	233.397	262.428	220.173	106.057	103.206	99.471
149	253.636	280.353	224.087	212.903	202.662	208.933	76.691	85.827	75.488
161	248.932	275.750	215.251	196.192	196.972	198.552	89.254	88.409	81.570
原厂软件									
5G	80M			40M			20M		
16#	TX+RX	TX	RX	TX+RX	TX	RX	TX+RX	TX	RX
36	246.972	273.766	221.075	228.917	241.359	221.069	117.831	121.367	105.262
149	248.812	278.817	213.213	217.126	208.853	209.783	118.105	124.807	102.637
161	248.532	280.977	221.166	182.389	198.742	176.069	106.702	108.708	89.450

5G	80M			20M		
原厂软件	TX+RX	TX	RX	TX+RX	TX	RX
27#	242.119	251.158	217.337	81.254	83.618	76.607
14#	242.639	260.362	224.990	83.514	83.415	77.552
11#	244.897	246.907	212.133	81.829	82.569	80.449
24#	234.666	240.054	217.414	98.616	98.124	92.855
磊科软件						
2#	229.412	200.712	199.022	110.266	105.580	106.680
22#	220.618	200.901	196.279	109.570	106.181	106.227
29#	218.284	201.563	198.549	109.265	104.926	106.568
12#	219.877	205.016	199.050	102.041	104.616	92.628
21#	230.012	205.165	201.563	109.619	105.992	107.404
17#	210.906	203.712	203.851	100.635	99.835	104.713

4.2、室内覆盖测试

1）测试环境:室温 25℃，笔记本电脑 2 台

2）参照物: PB-7268

3）陪测网卡:Realtek 8812AU

4）测试位置示意图

5）测试程序-IxChariot6.7

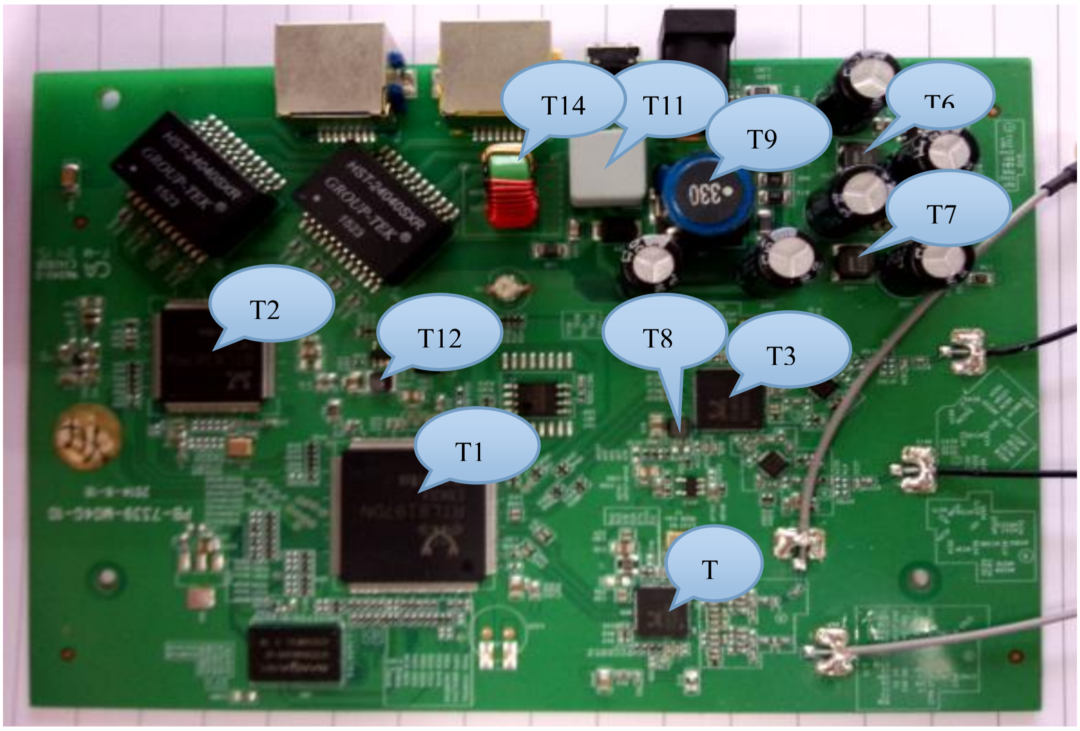
6）测试记录

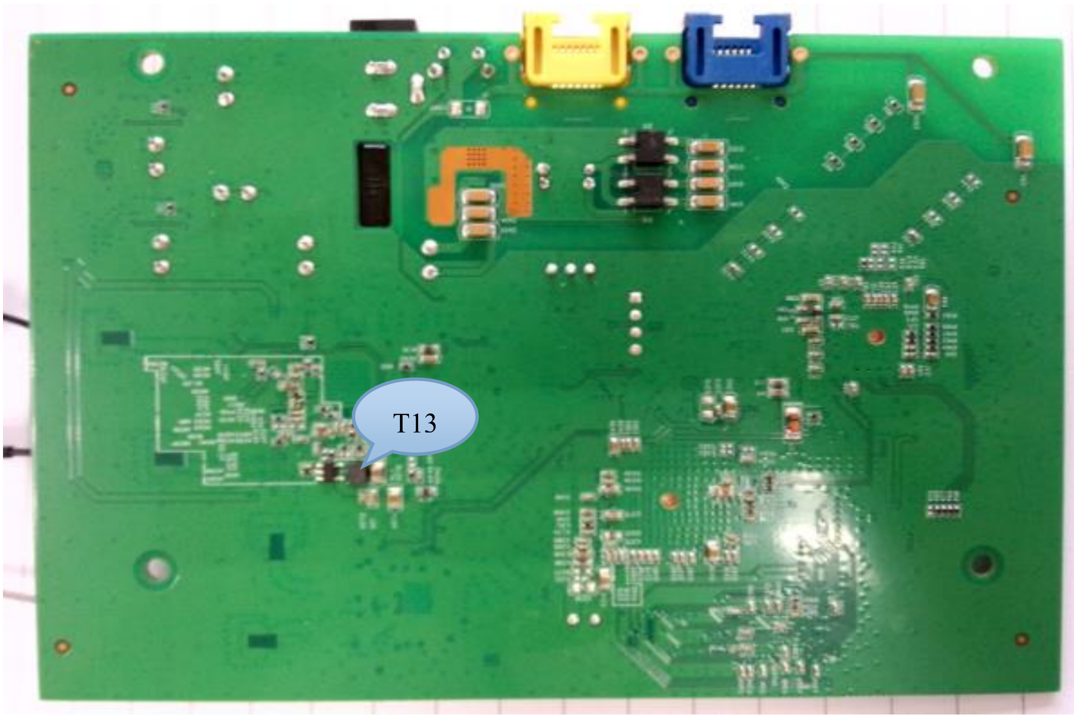
覆盖

	PB-7339		PB-7268		PB-7339		PB-7268	
带宽	2G		2G		5G		5G	
机型 测试点	20M	40M	20M	40M	20M	40M	20M	40M
A	12.062	4.121	6.282	5.210	41.113	74.999	13.893	14.115
B	35.741	41.699	16.093	17.868	91.379	92.649	30.643	55.968
C	26.010	19.812	19.603	28.205	81.285	93.928	38.646	51.008
D	6.850	28.769	3.512	13.781	65.590	89.734	43.045	54.662
E	11.525	6.090	1.767	6.063	65.663	70.483	39.880	32.982
F	9.825	14.402	5.582	9.758	43.715	63.850	30.162	41.494
G	10.549	12.103	1.907	9.425	50.866	66.080	39.856	45.843

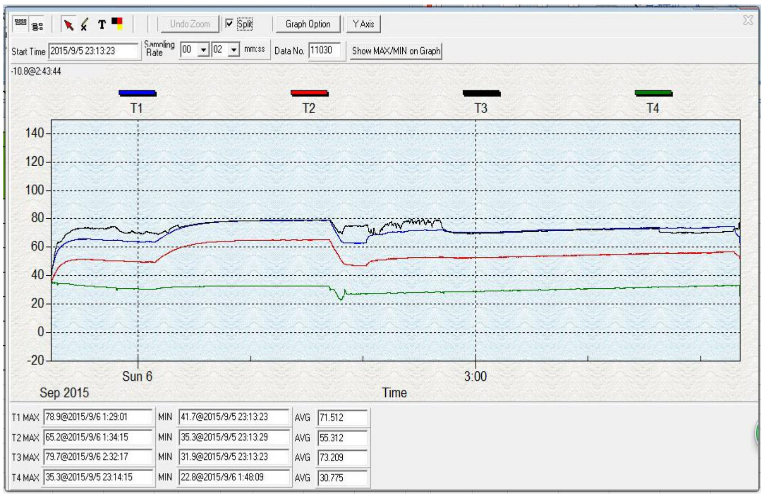
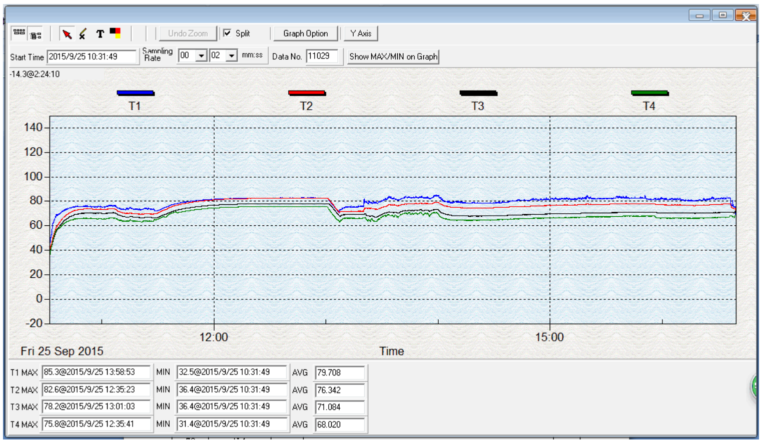
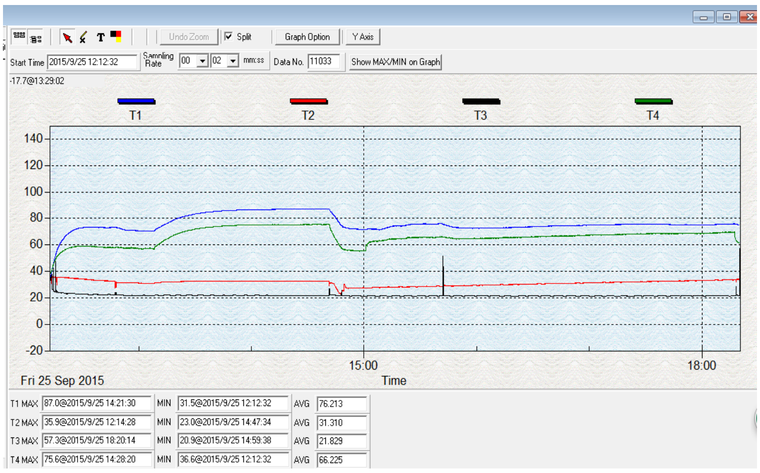
5、环境测试

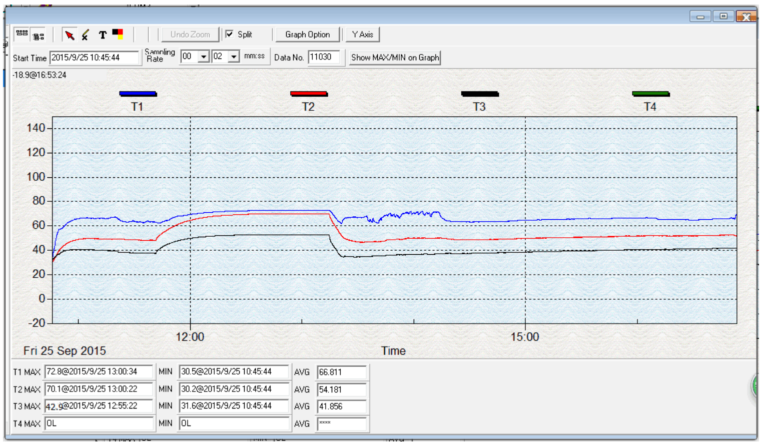
5.1、DUT 高低温功能及温度



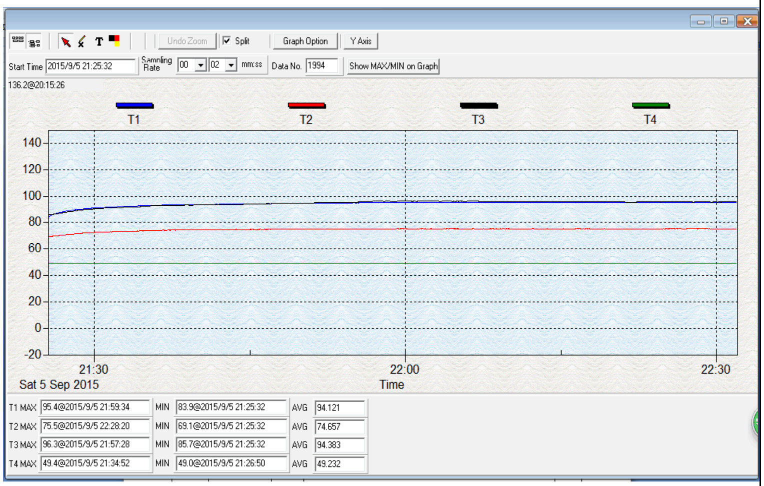
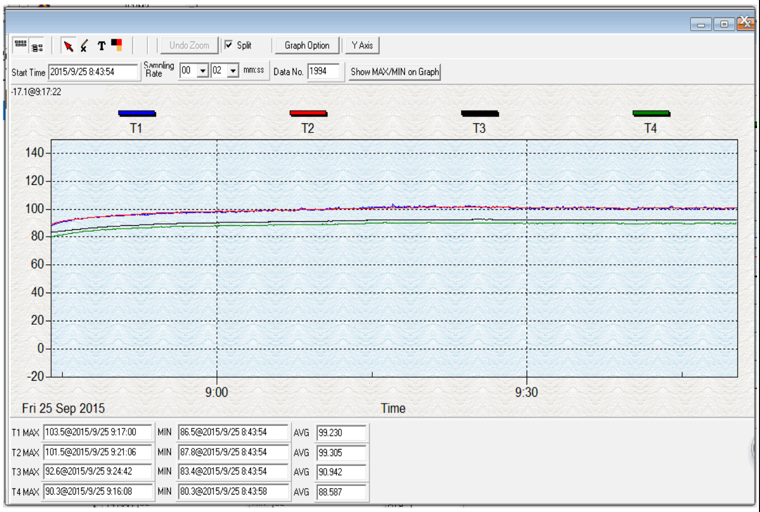


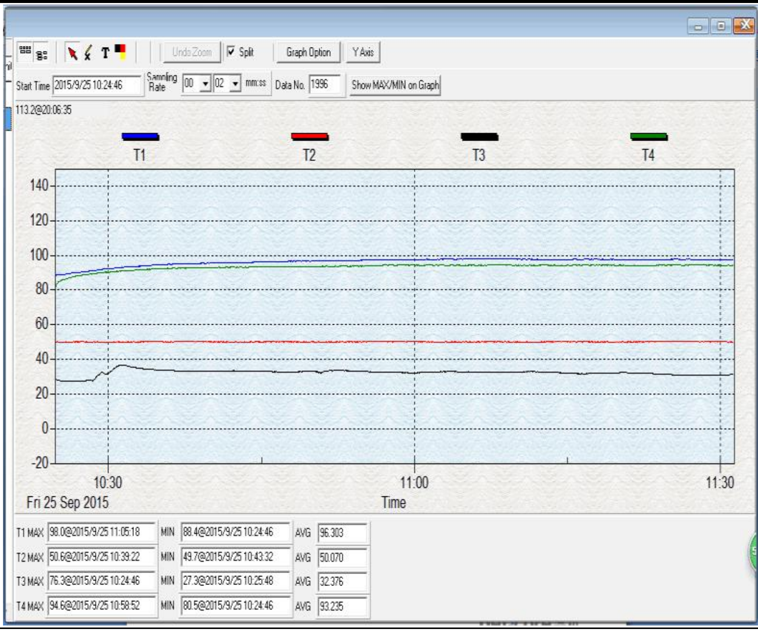
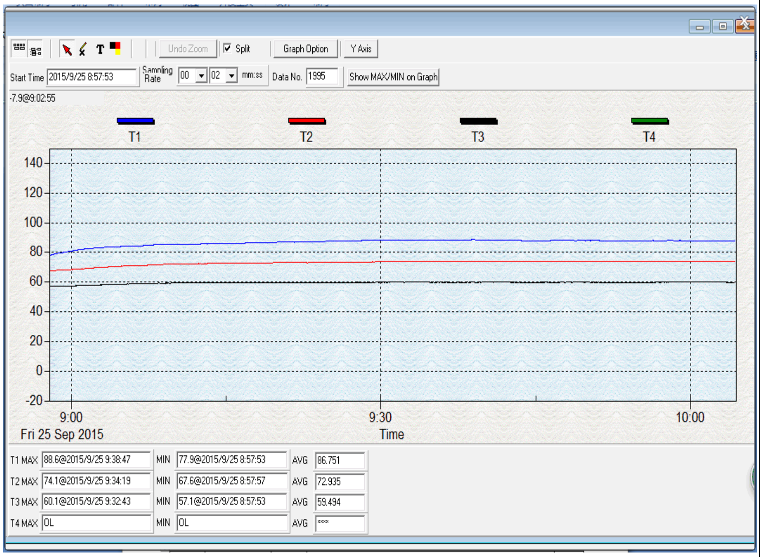
5. 1. 1、常温环境的温度

<table>
<tr><th>测试点编号</th><th>测试点</th><th>最高温度（℃）</th><th>温度曲线</th><th>外观有无异常</th></tr>
<tr><td>T1</td><td>U50（RTL8197DN）</td><td>78.4</td><td rowspan="4">
</td><td rowspan="4">无</td></tr>
<tr><td>T2</td><td>U4</td><td>65.2</td></tr>
<tr><td>T3</td><td>AU2</td><td>79.7</td></tr>
<tr><td>T4</td><td>环境</td><td>35.3</td></tr>
<tr><td>T5</td><td>U69</td><td>85.3</td><td rowspan="4">
</td><td></td></tr>
<tr><td>T6</td><td>LS11</td><td>82.6</td><td></td></tr>
<tr><td>T7</td><td>LS12</td><td>78.2</td><td></td></tr>
<tr><td>T8</td><td>AL1</td><td>75.8</td><td></td></tr>
<tr><td>T9</td><td>L32</td><td>87.0</td><td rowspan="4">
</td><td></td></tr>
<tr><td>T10</td><td>环境</td><td>35.9</td><td></td></tr>
<tr><td>T11</td><td>U8</td><td>57.3</td><td></td></tr>
<tr><td>T12</td><td>L934</td><td>75.6</td><td></td></tr>
</table>

<table>
<tr><td>T13</td><td>AL11</td><td>72.8</td><td rowspan="3">
</td><td></td></tr>
<tr><td>T14</td><td>LS13/LS14</td><td>70.1</td><td></td></tr>
<tr><td>T15</td><td>机壳</td><td>42.9</td><td></td></tr>
</table>

5.1.2、高温环境的温度

<table>
<tr><th>测试点编号</th><th>测试点</th><th>最高温度（℃）</th><th>温度曲线</th><th>外观有无异常</th></tr>
<tr><td>T1</td><td>U50（RTL8197DN）</td><td>95.4</td><td rowspan="4">
</td><td rowspan="4">无</td></tr>
<tr><td>T2</td><td>U4</td><td>75.5</td></tr>
<tr><td>T3</td><td>AU2</td><td>96.3</td></tr>
<tr><td>T4</td><td>环境</td><td>49.4</td></tr>
<tr><td>T5</td><td>U69</td><td>103.5</td><td rowspan="4">
</td><td></td></tr>
<tr><td>T6</td><td>LS11</td><td>101.5</td><td></td></tr>
<tr><td>T7</td><td>LS12</td><td>92.6</td><td></td></tr>
<tr><td>T8</td><td>AL1</td><td>90.3</td><td></td></tr>
</table>

<table>
<tr><td>T9</td><td>L32</td><td>98.0</td><td rowspan="4"></td><td></td></tr>
<tr><td>T10</td><td>环境</td><td>50.6</td><td></td></tr>
<tr><td>T11</td><td>U8</td><td>76.3</td><td></td></tr>
<tr><td>T12</td><td>L934</td><td>94.6</td><td></td></tr>
<tr><td>T13</td><td>AL11</td><td>88.6</td><td rowspan="3"></td><td></td></tr>
<tr><td>T14</td><td>LS13/LS14</td><td>74.1</td><td></td></tr>
<tr><td>T15</td><td>机壳</td><td>60.1</td><td></td></tr>
</table>

5.2、冷启动测试

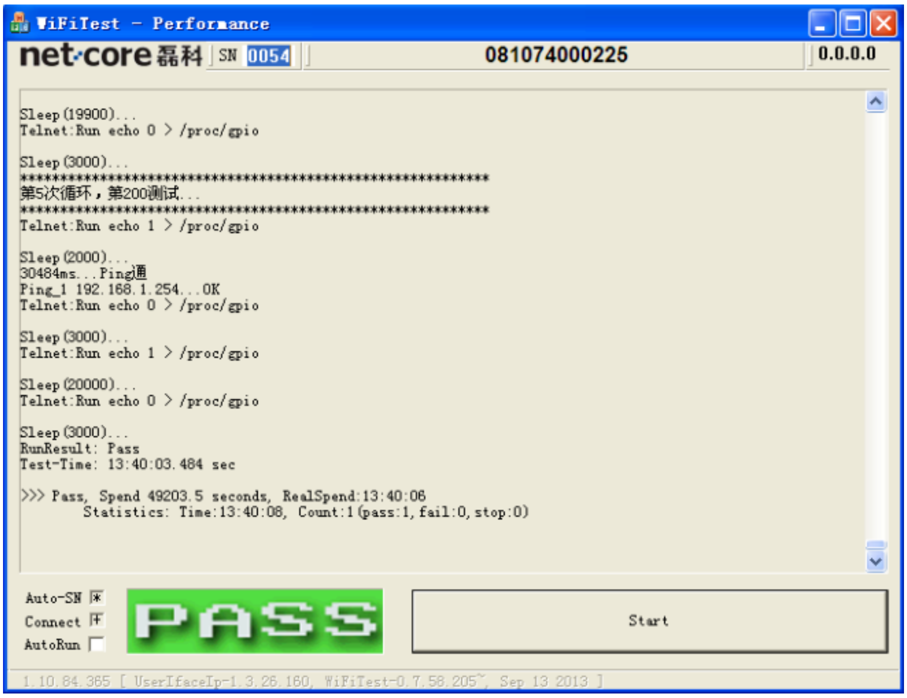
测试 OK

5.3、烧机

测试方法：在高温 50°环境下连续工作三天，两台 PC 通过 chariot 6.7 软件跑 throughout，查看长时间高温工作是否出现断线或者死机等现象。

测试结论：连续三天烧机未出现异常。

6、上下电测试

测试结果	
测试附件	Performance-Pass-SN0053-081074000225-20150923-105312.log